

Answer key

Version B

Marking guide and worked solutions · 50 points · **instructor copy: do not distribute before the exam**

How to mark this exam. Stages 1, 6 and parts of 2, 3, 4 and 5 assess reasoning, so there is no single right answer: give the mark when the choice is sensible **and defended with a reason or a number**. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in r , ± 0.2 in t and ± 0.5 in forecasts. The reference sheet fixes the exam's rules of thumb ($|z| > 2$ unusual, $|t| > 2$ significant, $|r| > 0.7$ heavy overlap between predictors); the course may use others, so change them in the key and on the sheet together. "Objective" gives the syllabus learning outcome (LO) and your category: **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1	LO1, LO5 · Con	Four variables, each with a sign and a mechanism. Plausible set: temperature (+, hot days want cold treats), weekend (+, more leisure time), beach_event (+, crowds nearby), price (–, law of demand). Also fine: sunshine (+), rainfall (–), humidity (–), post_change (either, with a reason).	4	1 per row: sign and a coherent mechanism. A "wrong" sign with a good reason scores; a sign with no reason does not.
2	LO2, LO5 · Con	Best answers: gelato_units (known only at closing time; caused by the same weather, so a symptom, not a cause) and iced_coffee_revenue_eur (price $\times$ units contains the outcome: circular). Also accept post_change (just a date flag that overlaps with price) or a weak-mechanism variable (wind) with a reason from how the café works.	2	1 per variable: must give a logical reason. "Not correlated" earns 0 (no data seen yet).
3	LO1 · Con	A result that contradicts the stated expectation: a coefficient with the opposite sign and $ t > 2$, or a coefficient whose $ t < 2$ (not distinguishable from zero).	1	1: names a coefficient/t-statistic outcome, not a vague "if it does not work".
4	LO2 · Con	(a) Validity: a mean over opening hours hides afternoon peaks and ignores humidity/wind, so it can misrepresent how hot it feels. (b) Reliability: sensor in sun versus shade, drift or recalibration changes readings with no change in weather.	2	1 validity + 1 reliability.
5	LO3 · Cal	(a) Rainfall $z = 6.89$; humidity $z = (95 - 66.58) \div 8.56 = 3.32$. (b) Rainfall: mean $0.76 >$ median 0.00 , right-skewed . Humidity: mean $66.58 \approx$ median 67.00 , roughly symmetric with one high value.	2	1 both z-scores + 1 shapes with the evidence.
6	LO3, LO4, LO5 · Con + Com	(a) 133 of 168 days (79%) are exactly 0, so rainfall does not separate most days; the distribution is strongly right-skewed (mean 0.76 , median 0 , max 14 , $z = 6.9$); in Figure 1 the dots stack at 0 and only a few wet days sit to the right, so the slope rests on few points; $r^2 = 0.05$. (b) The evidence weakens a little ($ t $ falls) but stays above 2, so the effect does not come only from the storm days; the skew is a warning that was checked, and here the result largely survives. (c) $=IF(E2>0,1,0)$. Accept equivalents such as $=IF(E2>=0.1,1,0)$ or $=--(E2>0)$.	3	1 explanation using Table 2/Figure 1 + 1 reading of the t-statistic change + 1 correct formula.
7	LO3, LO6 · Cal + Con	(a) $z = (56 - 112.79) \div 21.02 = -2.70$. (b) Keep it, with a caution. It follows the pattern (coldest day, fewest sales): Model 4 predicts 71, residual $-15 \approx -1.3$ SE; the slope and R^2 barely change. But it is the only day below 15°C , so the model has almost no evidence about cold days: do not extrapolate.	2	1 z + 1 reasoned decision (keep and the extrapolation caution, or a defended alternative).
8	LO3 · Com	(a) temperature_c, $r = 0.68$. (b) temperature_c and sunshine_hours ($r = 0.78$); iced_coffee_price_eur and post_change ($r = -0.77$). Both pairs are needed (note the negative sign for price and post_change).	2	1 (a) + 1 (b) both pairs.

9	LO3, LO5, LO6 · Con + Cal	(a) $r^2 = 0.78^2 = 0.61$: about 61% of the variation in sunshine moves with temperature; they largely measure the same thing (warm sunny days). (b) The model assumes predictors carry separate (independent) information. When two overlap, LINEST cannot tell which one is responsible: coefficients become unstable (they can shrink or flip sign), standard errors inflate and t-statistics fall even if the pair matters jointly. (c) Keep temperature (stronger r with iced coffee units: 0.68 vs 0.53; a direct mechanism for wanting a cold treat) and drop sunshine; keep iced_coffee_price_eur (varies daily including the random price trials, the lever the owner controls) and drop post_change (a date flag).	3	1 + 1 + 1. (c) needs a choice for each pair with a reason.
10	LO3, LO5 · Cal	$b = 0.68 \times 21.02 \div 3.43 = 4.17$; $a = 112.79 - 4.17 \times 22.80 = 17.8$; $\hat{y}(28) = 17.8 + 4.17 \times 28 = 134.4$ (accept ± 1 ; LINEST gives 134.6).	2	1 slope + 1 intercept and prediction.
11	LO6 · Cal + Con	(a) $t(\text{temp}, M1) = 4.19 \div 0.35 = 12.0$; $t(\text{temp}, M2) = 4.31 \div 0.56 = 7.7$; $t(\text{sunshine}, M2) = -0.21 \div 0.77 = -0.3$. (b) Adjusted R^2 : $M1 = 1 - (1 - 0.468) \times 167 \div 166 = 0.465$; $M2 = 1 - (1 - 0.468) \times 167 \div 165 = 0.462$ (it falls). (c) Sunshine adds almost no information beyond temperature (R^2 0.468 $\rightarrow$ 0.468) because the two overlap ($r = 0.78$). Temperature's SE rises from 0.35 to 0.56 and its t falls; sunshine's coefficient even changes sign. Drop sunshine.	3	1 three t-statistics + 1 two adjusted R^2 values + 1 explanation linking to the overlap.
12	LO5, LO6 · Com	(a) Row 1 is reversed: Col 1 = iced_coffee_price_eur (-17.47), Col 2 = beach_event (11.58), Col 3 = weekend (17.87), Col 4 = temperature_c (3.34), Col 5 = intercept (86.53). (b) $t = -17.47 \div 4.16 = -4.2$: significant; holding temperature, weekend and event constant, each €1 higher price goes with about 17 fewer iced coffees a day (a €0.20 change $\approx$ 3 units). (c) Any honest comparison with their own Stage 1 choices: supported expectations (temperature, weekend, event positive; price negative), or an expectation LINEST does not support, or an omitted variable that matters (price, event or weekend if left out).	3	1 all five matched + 1 t and interpretation "holding constant" + 1 look-back that names a specific expectation.
13	LO3, LO6, LO7 · Cal + Con	(a) $SE = \sqrt{(16.4^2/84 + 17.8^2/84)} = 2.64$; difference = 24.5; $t = 9.26$: significant. (b) Simple slope = $-0.53 \times 21.02 \div 0.25 = -44.7$ units per €; Model 4: -17.47 per €. A €0.40 cut: 17.9 extra units (simple) vs 7.0 (Model 4); observed +24.5. Model 4 credits the cut with about 29% of the jump. (c) Trust Model 4: it holds temperature, weekend and event fixed. The cut coincided with summer (price and temperature correlate at -0.50), so the simple slope credits price with the weather's effect (confounding).	3	1 SE and t + 1 two slopes and unit effects + 1 choice with the confounding reason.
14	LO6 · Cal	May 1: predicted 108.8, residual -0.8, $ res < 22.7$: ordinary. Aug 20: predicted 130.1, residual -29.1, $ res > 22.7$: unusual . Action: investigate (stock-out, closure, forecast error, recording error); do not delete without a reason. About 1 day in 20 beyond 2 SE is expected even if the model is fine.	2	1 two predictions and residuals + 1 correct classification and sensible action.
15	LO4, LO6 · Con	(a) Add weekend first, with a positive sign: residual means +12.8 (weekend) vs -5.1 (weekday), a gap of 17.9, the largest; in Figure 2 the red weekend dots sit above 0 and the hollow weekday dots below. Model 1 under-predicts weekends and over-predicts weekdays. Next: beach_event (gap 13.2), then price (negative; gap 9.9). (b) Dots scattered randomly around the zero line, no pattern, weekend and weekday dots mixed, similar spread at every temperature.	2	1 variable + sign + evidence from the table or figure + 1 sketch showing a patternless cloud around 0.
16	LO5, LO6 · Cal	Baseline: Aug 8 131.2, Aug 13 110.2, Aug 25 107.6, Sep 2 111.9. [errors]: baseline 6.8, 12.8, 10.4, 17.1 $\rightarrow$ MAE 11.8 ; M4 9.5, 7.7, 3.2, 1.5 $\rightarrow$ MAE 5.5 ; M5 10.7, 4.8, 3.3, 3.3 $\rightarrow$ MAE 5.5 .	5	1 baseline forecasts + 2 errors (≥ 10 of 12 correct = 2, 7-9 = 1) + 2 MAEs (all three = 2, two = 1).
17	LO6 · Con	(a) R^2 can only rise when variables are added: it measures fit to the days already seen, noise included, so it cannot show better prediction of new days. On the unseen days M5 is no better (5.5 vs 5.5 here; 9.8 vs 9.7 over all 56 unseen days). (b) Give the owner M4 : the same accuracy with half the inputs. It needs only the temperature forecast, the calendar and the café's own price; M5 needs four more weather forecasts that overlap with temperature, are themselves uncertain, and are harder to explain. The baseline (MAE 13.1 over all 56 days) is much worse than either model. A student who picks M5 must defend it with unseen-day evidence, which does not support it here.	2	1 in-sample vs unseen reasoning + 1 choice defended with accuracy and with what must be known in advance.

18	LO7 · Con	Plain-language memo: recommends temperature, weekend, beach_event and price (with why); gives a forecast error in cups (SE of $y \approx 11$ for M4, or $MAE \approx 10$ on unseen days); one limitation (a single April–September season of synthetic data; weather forecast error; association is not proof of cause; the model cannot foresee unusual days; do not extrapolate beyond the observed range).	3	1 recommendation with a reason + 1 a number in plain words + 1 a genuine limitation.
19	LO1, LO5, LO7 · Con	Open response. Strong answers name a plausible variable not in the table (e.g. forecast crowd size, school holidays, competitor price, cruise-ship arrivals, a heat-wave flag), say how it is measured, give a sign with a mechanism, propose adding it to LINEST (t-statistic, SE of y, adjusted R^2) and testing on unseen days (MAE), and say what result would count against it.	4	1 variable and measurement + 1 sign and mechanism + 1 testing plan + 1 falsifier.

2 · Worked tables

Q16, hold-out days (Table 8). Baseline = $0.5 \times (1 \text{ week ago}) + 0.3 \times (2 \text{ weeks ago}) + 0.2 \times (3 \text{ weeks ago})$.

Day	Actual	Baseline calculation	Baseline	M4	M5	errors (base / M4 / M5)
Aug 8	138	$0.5 \times 120 + 0.3 \times 146 + 0.2 \times 137$	131.2	128.5	127.3	6.8 / 9.5 / 10.7
Aug 13	123	$0.5 \times 108 + 0.3 \times 108 + 0.2 \times 119$	110.2	115.3	118.2	12.8 / 7.7 / 4.8
Aug 25	118	$0.5 \times 96 + 0.3 \times 116 + 0.2 \times 124$	107.6	121.2	121.3	10.4 / 3.2 / 3.3
Sep 2	129	$0.5 \times 103 + 0.3 \times 120 + 0.2 \times 122$	111.9	127.5	125.7	17.1 / 1.5 / 3.3
MAE						11.8 / 5.5 / 5.5

Q14, predictions from Model 4.

Day	Calculation	Predicted	Residual
May 1	$86.53 + (3.34 \times 23.9) + (17.87 \times 0) + (11.58 \times 0) + (-17.47 \times 3.3)$	108.8	-0.8
Aug 20	$86.53 + (3.34 \times 28.2) + (17.87 \times 0) + (11.58 \times 0) + (-17.47 \times 2.9)$	130.1	-29.1

3 · Where students go wrong

- **Reading LINEST Row 1 in the order of the X columns** (Q12). It is reversed, so the first number is the last X column. Give no mark for part (a) if the order is not reversed, but apply follow-through to (b).
- **Treating $|r| < 0.7$ as “no overlap” and ignoring negative correlations** (Q8). A pair at -0.75 counts.
- **“Delete the outlier”** (Q7). A raw extreme is not an error: ask whether the model explains it. Mark the reasoning, not the decision.
- **Dropping price because $r \approx 0$** (Q13c, Q1) or **trusting the simple price slope** (Q13b): both ignore confounding.
- **“M5 is best because its R^2 is higher”** (Q17). Look for the argument that R^2 only rises with more variables and that what matters is unseen days.
- **Hypothesis marks:** never penalise a hypothesis because the data disagree; penalise a hypothesis that no data could disagree with.